Example 9.4

Suppose that the MA(1) model

$$X_t = Z_t' + 0.6 Z_{t-1}' \quad (1)$$

is fitted to a set of time series data.

The residuals $\{Z_t'\}$ are such that

$$Z_t' = (1 - 0.8B) Z_t \quad (2)$$

In (1)

$$X_t = (1 + 0.6B) Z_t'$$

Subs. (2) into the above

$$\Rightarrow \quad X_t = (1 + 0.6B)(1 - 0.8B) Z_t$$

$$= (1 - 0.2B - 0.48B^2) Z_t$$

$$= Z_t - 0.2 Z_{t-1} - 0.48 Z_{t-2}.$$